

PetPlants — Product Specification

1. Overview

PetPlants is a gamified plant care web app for two users sharing the same account (no login required). One user sets up plant profiles and schedules remotely; the other waters plants and logs check-ins at home. All data is shared in real time via a backend database.

Tagline: *Your plants are your pets. Treat them like it.*

2. Design System

2.1 Visual Style: Pixel Art + Cozy Game

Think **Stardew Valley meets iOS Health**. The aesthetic is warm, pixel-art-inspired, cozy, and **dark-only** — no light mode.

Token	Value
Primary background	#0d0d1a (deep space dark)
Surface / card	#1a1a2e
Card border	#2a2a4a
Accent green	#00ff87 (neon pixel green)
Accent yellow	#ffd166
Accent pink	#ff6b9d
Accent blue	#6bcbff
XP gold	#ffcc00
Text primary	#e8f4f8
Text muted	#556080

Token	Value
Border radius	0px for pixel-art cards and buttons; 2px for inputs
Pixel grid	All icons and illustrations use 16×16 or 32×32 base grids
Image rendering	image-rendering: pixelated on all sprites/SVGs

2.2 Typography

Use **both** of these Google Fonts — load via `<link>` in `index.html`:

```
<link
href="https://fonts.googleapis.com/css2?family=Press+Start+2P&family=VT323:wght@400&display=swap" rel="stylesheet">
```

Use	Font	Size guidance
App title, section headers, level badge	Press Start 2P	10–14px (this font is large at small sizes)
Plant names on cards, nav labels	Press Start 2P	8px
Body text, notes, dates, descriptions	VT323	18–22px (VT323 needs larger size to be readable)
XP numbers, stat readouts	VT323	24px

Important for developers: Press Start 2P at 12px already looks like a headline. Never go above 16px for it. VT323 looks blurry below 16px — use 18px minimum. Line height for Press Start 2P should be 1.8 due to its tight descenders.

2.3 Plant Emoticons (Animated)

Four pixel-art plant styles: **Leafy**, **Succulent**, **Flower**, **Herb**. Each style has 4 emotional states rendered as CSS-animated SVG sprites (32×32px base, scaled 2× or 3× in UI).

State	Trigger	Animation
😊 Happy	Watered within schedule	Gentle leaf sway loop (CSS <code>rotate ±3deg</code> , 2s ease-in-out infinite)
😓 Thirsty	Overdue for watering	Slow droop + water drop tear falling (translate Y, fade out, loop)
😴 Resting	Watered today	Eyes closed, Zzz floats up and fades (3 staggered Zzz elements)
☁️ Postponed	Shifted due to moist soil	Small pixel rain cloud above plant, rain drops falling

All animations use `@keyframes`. No JavaScript animation libraries. Keep `will-change: transform` on animated elements for GPU compositing.

2.4 Layout

- **Mobile-first**, single-column, max content width `420px`, centered on desktop with dark background extending full width
- **Bottom nav bar** (fixed, pixel-art icons): Dashboard · Plants · Log
- Cards: `1px solid #2a2a4a` border, subtle `box-shadow: 0 0 12px rgba(0,255,135,0.08)` green glow
- Buttons: pixel-art style — no border-radius, `2px solid` accent color border, hover shifts background to accent color
- Inputs: dark background (`#0d0d1a`), `1px solid #2a2a4a`, green focus border `#00ff87`

3. Pages & Features

3.1 Dashboard (Home)

The main page. Shows game state, XP progress, and all plants at a glance.

Components (top to bottom):

1. **App header:** **PetPlants** in **Press Start 2P**, green, centered. Small pixel leaf icon beside it.
2. **Player level banner:** Full-width card showing current level, title, XP bar (see Section 3.7)
3. **Today's summary strip:** e.g. *"3 plants need water today"* in accent yellow, **VT323** font
4. **Plant cards grid:** 2-column grid of plant cards
5. Plants with "Water today!" status float to the top of the grid

Plant Card:

- Animated plant emoticon (top center of card)
 - Plant name in **Press Start 2P** at 8px
 - Next watering label in **VT323**
 - "Water today!" pill badge in accent pink if due
 - Soil moisture badge if logged today: **Moist** 💧 or **Dry**
 - Tap → goes to plant detail page
-

3.2 Plant Profile Page

Displays:

- Large plant emoticon (3× scale, animated)
 - Plant name + species in **Press Start 2P**
 - Watering schedule: *"Every 3 days · 200ml"* in **VT323**
 - Last watered date
 - Notes (editable inline or via Edit button)
 - Edit button → opens Edit Plant form
 - Full 10-day history bar (see Section 3.4)
 - XP earned from this plant (e.g. *"This plant has earned you 240 XP"*)
-

3.3 Add / Edit Plant Form

Fields:

Field	Type	Notes
Plant name	Text input	Required
Plant type / species	Text input	Optional

Field	Type	Notes
Watering frequency	Number (days)	e.g. 3 = every 3 days. Required.
Water amount	Text input	e.g. 200ml or 1 cup. Optional.
Notes	Textarea	Weather notes, placement, etc. Optional.
Emoticon style	Visual selector	4 pixel plant styles: Leafy · Succulent · Flower · Herb. Tap to select, shows preview of Happy state.

Validation: Name and frequency required. Show inline pixel-art error messages (red border + small ! icon).

3.4 10-Day History Tracker

Inspired by iOS Health activity bars. Shown compact on Dashboard inside each plant card, and full-size on Plant Detail page.

Design:

- 10 columns (D-9 to Today), each a pixel bar 8px wide, 32px tall on compact / 48px tall on full view
 - Bar fill colors:
 - #00ff87 green = watered
 - #2a2a4a empty = not due / no data
 - #ffd166 yellow = due but not watered (missed)
 - #6bcbff blue = postponed (soil moist)
 - Day label below each bar in VT323 12px: M, T, W, etc.
 - Today's bar gets a box-shadow: 0 0 6px #00ff87 glow
 - On hover/tap, show tooltip: date + status
-

3.5 Watering Log Page

Accessible via bottom nav "Log" tab or "Water today!" tap on a plant card.

Flow:

1. List of plants due today (or all plants if none due)
2. Each plant shows emoticon + name + a two-button row:
 -  **Watered** (green button)
 -  **Moist — skip today** (blue button)
3. On confirm:
 - Pixel confetti burst animation (CSS keyframes, coloured squares scatter from center)
 - XP gain toast notification pops up: **+25 XP**  in **Press Start 2P**
 - Plant emoticon switches to Resting state

Under 3 taps. No extra fields.

3.6 Smart Watering Shift

When user taps "Moist — skip today":

- Next watering date shifts +1 day
- Status logged as **postponed**
- Plant emoticon → Postponed state
- Auto-note: *"Shifted — soil moist on [date]"*
- Awards **5 XP** (less than watering, but rewards engagement)

Can be triggered multiple consecutive days.

3.7 XP & Levelling System

Gamifies consistency. Shown on Dashboard in a persistent level banner.

XP Rules:

Action	XP Gained
Water a plant on schedule	+25 XP
Water a plant (1 day late)	+15 XP

Action	XP Gained
Water a plant (2+ days late)	+5 XP
Log "soil moist" check-in	+5 XP
Add a new plant	+10 XP
3-day watering streak (any plant)	+20 XP bonus
7-day streak	+50 XP bonus

Level thresholds:

Level	XP Required	Title
1	0	Seedling 🌱
2	100	Sprout 🌿
3	250	Grower 🍀
4	500	Tender 🌸
5	900	Botanist 🌺
6	1400	Garden Keeper 🌳
7	2000	Plant Whisperer 🌲
8	3000	Master Gardener 🏡

Level Banner UI (on Dashboard):

- Full-width card, dark surface
- Left: pixel plant icon matching current level tier
- Center: Level number in **Press Start 2P** + title in VT323
- Bottom: XP progress bar — pixel-art style, fills left to right in gold (#ffcc00), shows [current XP] / [next level XP] in VT323
- On level-up: full-screen pixel confetti + level-up text animation (**LEVEL UP!** bounces in **Press Start 2P**)

XP is global (not per-plant), stored in a single **player_stats** table.

4. Data Model

Plants Table

```
CREATE TABLE plants (  
  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  
  name TEXT NOT NULL,  
  
  species TEXT,  
  
  frequency_days INTEGER NOT NULL,  
  
  water_amount TEXT,  
  
  notes TEXT,  
  
  emoticon_style TEXT DEFAULT 'leafy' CHECK(emoticon_style IN  
('leafy', 'succulent', 'flower', 'herb')),  
  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
  
);
```

Watering Log Table

```
CREATE TABLE watering_log (  
  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  
  plant_id INTEGER REFERENCES plants(id) ON DELETE CASCADE,  
  
  log_date DATE NOT NULL,  
  
  status TEXT CHECK(status IN ('watered', 'postponed', 'missed')),  
  
  notes TEXT,  
  
  xp_awarded INTEGER DEFAULT 0,
```



```
created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
  
);
```

Player Stats Table

```
CREATE TABLE player_stats (  
  
  id INTEGER PRIMARY KEY DEFAULT 1,  
  
  total_xp INTEGER DEFAULT 0,  
  
  current_level INTEGER DEFAULT 1,  
  
  current_streak INTEGER DEFAULT 0,  
  
  longest_streak INTEGER DEFAULT 0,  
  
  last_action_date DATE  
  
);
```

Seed with one row: `INSERT INTO player_stats (id) VALUES (1);` — only ever one row.

Next Watering Date (computed, not stored)

- Start from last `watered` entry date for the plant
- Add `frequency_days`
- Add 1 day for each `postponed` entry logged after that watered date

5. Tech Stack

Layer	Choice	Reason
Frontend	React (Vite)	Component-based, good for animated emoticons + XP bar
Styling	Plain CSS + CSS variables	No build complexity, pixel-art friendly

Layer	Choice	Reason
Backend	Node.js + Express	Simple REST API
Database	SQLite via better-sqlite3	Zero config on Replit, persistent
Animations	CSS @keyframes only	No animation libraries needed
Fonts	Press Start 2P + VT323 via Google Fonts	Authentic pixel look
Hosting	Replit (free tier)	~10–30s cold start acceptable

6. API Endpoints

GET	/api/plants	— all plants + computed next-water date + current state
POST	/api/plants	— add plant (awards 10 XP)
PUT	/api/plants/:id	— edit plant
DELETE	/api/plants/:id	— delete plant
GET	/api/plants/:id/history	— last 10 days log for a plant
POST	/api/log	— log watering action (watered / postponed), returns XP awarded
GET	/api/dashboard	— all plants + today summary + player stats
GET	/api/stats	— player XP, level, streak

7. Replit Project Setup

1. Create a new Replit — choose **Node.js** template
2. Project structure:

/

└─ server/

| └─ index.js

| └─ db.js (SQLite init, schema, seed player_stats)

| └─ xp.js (XP calculation logic)

| └─ routes/

| └─ plants.js

| └─ log.js

| └─ stats.js

└─ client/

| └─ index.html (Google Fonts link goes here)

| └─ src/

| └─ App.jsx

| └─ pages/

| | └─ Dashboard.jsx

| | └─ PlantDetail.jsx

| | └─ AddEditPlant.jsx

| | └─ Log.jsx

| └─ components/

| | └─ PlantCard.jsx

| | └─ PlantEmoticon.jsx (SVG sprites per style+state)

| | └─ HistoryBar.jsx

```

|   |   | — XPBanner.jsx
|   |   | — XPToast.jsx
|   |   | — BottomNav.jsx
|   |   | — styles/
|       | — global.css      (CSS variables, font imports, base styles)
| — package.json
| — .replit

```

3. `.replit` run command:

```
npm run build && node server/index.js
```

4. Express serves built React app as static from `/client/dist`

5. `petplants.db` lives in project root — Replit persists this across restarts

8. Out of Scope (v1)

- Push / email notifications
- User authentication / multi-account
- Automatic weather API integration
- Plant image uploads
- Fertilizer or repotting tracking
- Social / leaderboard features

9. Decisions Confirmed

- ☒ **Emoticon styles:** Leafy, Succulent, Flower, Herb
- ☒ **Theme:** Dark only — no toggle
- ☒ **Missed days:** Yellow bar only, no additional warning
- ☒ **Fonts:** `Press Start 2P` (headings, labels) + `VT323` (body, stats)
- ☒ **XP system:** Global player XP, 8 levels, streak bonuses